

DATA QUALITY REPORT

Project FORESIGHT — Demand & Inventory Intelligence

Client: NorthBay Living | Zidio Development Internship

Dataset: retail_forecasting_cleaned_v2 | August 2026

1. Source Data Overview

Item	Detail
Original file	Master Data.xlsx (denormalized extract)
Grain	Daily (SKU × Date)
Rows	365,000
Date range	2021-08-14 → 2026-08-12 (~5 years)
Unique SKUs	200
Categories	Decor, Small Appliances, Furniture

The raw extract was a single denormalized table combining sales, SKU master, inventory, and calendar attributes. It contained missing values, inconsistent calendar fields, and minor logical inconsistencies typical of real-world client extracts.

2. Issues Found & How They Were Handled

2.1 Junk / Unnamed Columns

Issue: Two trailing columns (Unnamed: 20, Unnamed: 21) contained no useful data.

Action: Dropped.

Rationale: Not part of the business schema.

2.2 Missing unit_price (1,095 rows)

Issue: unit_price was null on some records.

Investigation: When units_sold > 0, price could be recovered as revenue / units_sold. When units_sold = 0, no transaction price existed.

Action: (1) Filled with revenue / units_sold where possible. (2) Remaining nulls filled with list_price.

Result: 0 missing unit_price in final dataset.

Rationale: Best available proxy for actual selling price; list_price is a reasonable fallback when no sale occurred.

2.3 Missing on_order_units (730 → 143 remaining)

Issue: 730 nulls in on_order_units.

Investigation: Examined previous and next values per SKU using forward/backward fill logic.

Action: Filled only when previous value equalled next value (safe interpolation). Left 143 ambiguous cases as null.

Rationale: Avoid inventing inventory positions that cannot be justified. The 143 rows are documented and can be treated as 0 or excluded in modelling if needed.

2.4 Broken Calendar Fields (week, month, season)

Issue: Original week, month, and season matched the actual date in only ~2–8% of rows. Values were essentially random.

Action: Regenerated from the real date column — month = date.month, week = ISO week number, season mapped as Winter (12–2), Spring (3–5), Summer (6–8), Fall (9–11).

Result: 100% consistency with actual date.

Rationale: Seasonality and time-based features are critical for forecasting. Using the original fields would have produced incorrect models.

2.5 promo_event Missing Values

Issue: Large number of nulls in promo_event.

Action: promo_flag = False → “No Promotion Recorded”; promo_flag = True → “Unknown Promotion”.

Rationale: Preserves the signal that a promotion occurred even when the event name was missing.

2.6 Pre-launch Records (36,937 rows)

Issue: Rows where date < launch_date (10.1% of data).

Investigation: All of these rows had units_sold = 0.

Action: Kept in the dataset but flagged for awareness. Can be filtered for modelling if desired.

Rationale: No actual sales occurred before launch, so they do not distort demand patterns.

2.7 Revenue Consistency & Other Checks

Revenue: Small differences (max 0.42) between revenue and units_sold × unit_price on ~29k rows — consistent with floating-point rounding. No change made.

Duplicates: 0 full-row duplicates and 0 date+sku duplicates.

Negatives: 0 negative values in any numeric column.

3. Final Cleaned Dataset Summary

Metric	Value
File	retail_forecasting_cleaned_v2
Rows × Columns	365,000 × 22
Missing values	Only 143 in on_order_units
Calendar consistency	100%
Duplicates	0
Negative values	0
Stockout flag (is_stockout)	Added — 9.41% of days

4. Remaining Known Limitations

1. **143 missing on_order_units** — left as null by design.
2. **Pre-launch rows** still present (zero sales only).
3. **region column** appears in the cleaned file; origin should be confirmed with the data provider.
4. Analysis is based on the provided historical extract only — no live system integration.

5. Impact on Downstream Work

- Weekly aggregation and forecasting can now proceed with reliable calendar features.
- Dead-stock and stockout analyses are possible using the cleaned inventory fields.

- The dataset is reproducible from the raw extract once the cleaning steps are finalised into a single pipeline script.

6. Conclusion

The data is now analysis-ready. All major quality issues have been investigated, documented, and either corrected or explicitly left with a clear rationale. The remaining limitations are minor and do not block demand forecasting or risk scoring. This report satisfies the acceptance criteria for Deliverable D1 (Data Pipeline) and the data-quality portion of Deliverable D2 (Data-quality & EDA insight memo).